

Security of the Stream Cipher from Pseudorandomness of its Generator

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Abstract

We give a self-contained, reductionist proof that the canonical stream-cipher encryption scheme—encryption by XOR with the output of a pseudorandom generator (PRG)—achieves indistinguishability of encryptions in the presence of an eavesdropper, provided the underlying generator is a secure PRG. The argument is presented in prose, with the single reduction algorithm displayed as pseudocode. The reduction is advantage-preserving: the eavesdropping advantage against the cipher is bounded above by the distinguishing advantage against the generator, with no loss. We isolate the one-time-pad as the information-theoretic anchor of the hybrid and remark on tightness, on the necessity of the assumption, and on the failure of the single-message guarantee to extend to the multi-message setting for the stateless scheme.

1 Preliminaries and notation

We work in the asymptotic (uniform-complexity) model of computation. Throughout, $n \in \mathbb{N}$ denotes the security parameter, and all algorithms receive 1^n as (implicit or explicit) input. We write $x \xleftarrow{\$} S$ for the assignment to x of an element drawn uniformly at random from a finite set S , and $y \leftarrow \mathcal{A}(x)$ for the assignment to y of the (possibly randomized) output of an algorithm $\mathcal{A}$ on input x . By PPT we mean probabilistic polynomial-time (in n).

Definition 1 (Negligible function). A function $\text{negl}: \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible* if for every positive polynomial p there exists $N \in \mathbb{N}$ such that $\text{negl}(n) < 1/p(n)$ for all $n > N$. Equivalently, $\text{negl}(n) = n^{-\omega(1)}$.

The class of negligible functions is closed under addition and under multiplication by any fixed polynomial; we use both facts without comment.

For a bit string x we write $|x|$ for its length and $\oplus$ for bitwise exclusive-or; for equal-length strings x, y the map $z \mapsto z \oplus x$ is a bijection on $\{0, 1\}^{|x|}$, a fact we invoke repeatedly.

2 The two definitions in play

2.1 Pseudorandom generators

Let $\ell: \mathbb{N} \rightarrow \mathbb{N}$ be a polynomial with $\ell(n) > n$ for all n ; ℓ is the *expansion factor*.

Definition 2 (Pseudorandom generator). A deterministic polynomial-time computable function $G: \{0, 1\}^* \rightarrow \{0, 1\}^*$ with $|G(s)| = \ell(|s|)$ for all s is a *pseudorandom generator* with expansion ℓ if

for every PPT distinguisher $\mathcal{D}$ the *distinguishing advantage*

$$\mathbf{Adv}_{G,\mathcal{D}}^{\text{prg}}(n) := \left| \Pr_{s \xleftarrow{\$} \{0,1\}^n} [\mathcal{D}(G(s)) = 1] - \Pr_{r \xleftarrow{\$} \{0,1\}^{\ell(n)}} [\mathcal{D}(r) = 1] \right|$$

is negligible in n . Both probabilities are additionally over the internal randomness of $\mathcal{D}$.

The seed length n is exactly the security parameter, so a uniform seed is an n -bit string and its image is an $\ell(n)$ -bit string; the gap $\ell(n) - n$ is the number of bits by which G “stretches” its input. It is precisely this stretch that makes the definition nontrivial: were $\ell(n) = n$, the identity function would satisfy Definition 2.

2.2 Private-key encryption and eavesdropping security

A *private-key encryption scheme* is a triple $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ of PPT algorithms with an associated message space $\{\mathcal{M}_n\}_n$, satisfying the usual correctness condition: for every n , every key k in the support of $\text{Gen}(1^n)$, and every $m \in \mathcal{M}_n$, we have $\text{Dec}_k(\text{Enc}_k(m)) = m$ with probability 1.

We adopt the standard indistinguishability formulation of eavesdropping security. For a scheme Π , an adversary $\mathcal{A}$, and a value n , define the experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n)$ as follows.

1. $\mathcal{A}(1^n)$ outputs a pair of messages $m_0, m_1 \in \mathcal{M}_n$ with $|m_0| = |m_1|$.
2. A key is generated, $k \leftarrow \text{Gen}(1^n)$, a challenge bit is chosen, $b \xleftarrow{\$} \{0,1\}$, and the *challenge ciphertext* $c \leftarrow \text{Enc}_k(m_b)$ is computed and handed to $\mathcal{A}$.
3. $\mathcal{A}$ outputs a bit b' . The experiment *evaluates to 1* (“ $\mathcal{A}$ succeeds”) if $b' = b$, and to 0 otherwise.

Definition 3 (Indistinguishable encryptions under eavesdropping). Π has *indistinguishable encryptions in the presence of an eavesdropper*, or is *eav-secure*, if for every PPT adversary $\mathcal{A}$ the *eavesdropping advantage*

$$\mathbf{Adv}_{\Pi,\mathcal{A}}^{\text{eav}}(n) := \Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] - \frac{1}{2}$$

is negligible in n .

The subtracted $1/2$ is the success probability of the trivial adversary that ignores c and guesses; eav-security asks that no efficient strategy beat this baseline by a non-negligible margin. We record for later use the information-theoretic gold standard against which the computational notion is calibrated.

Definition 4 (Perfect secrecy). Π is *perfectly secret* if for every n , every adversary $\mathcal{A}$ whatsoever (of arbitrary computational power), and the experiment above, $\Pr[\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}(n) = 1] = \frac{1}{2}$ exactly.

The relevant instance is Vernam’s one-time pad, whose perfect secrecy we invoke as a lemma.

Lemma 1 (One-time pad). *Fix n and let $L := \ell(n)$. Let $\tilde{\Pi}$ denote the scheme with message space $\{0,1\}^L$ in which Gen outputs a uniform pad $p \xleftarrow{\$} \{0,1\}^L$ and $\text{Enc}_p(m) = p \oplus m$. Then $\tilde{\Pi}$ is perfectly secret; in particular $\Pr[\text{PrivK}_{\mathcal{A},\tilde{\Pi}}^{\text{eav}}(n) = 1] = \frac{1}{2}$ for every $\mathcal{A}$.*

Proof. Condition on any fixed messages $m_0, m_1 \in \{0,1\}^L$ output by $\mathcal{A}$ and on any fixed challenge bit b . Because p is uniform on $\{0,1\}^L$ and independent of b , and $x \mapsto x \oplus m_b$ is a bijection of $\{0,1\}^L$, the challenge $c = p \oplus m_b$ is uniform on $\{0,1\}^L$; crucially, its distribution does not depend on b . Hence the view of $\mathcal{A}$ is statistically independent of b , and $\mathcal{A}$ ’s output b' is independent of b . Averaging over the uniform choice of $b \in \{0,1\}$ gives $\Pr[b' = b] = \frac{1}{2}$. $\square$

Algorithm 1 Distinguisher $\mathcal{D}$ against G , with black-box access to eavesdropper $\mathcal{A}$

Require: challenge string $w \in \{0, 1\}^{\ell(n)}$ (equal to $G(s)$ or to uniform r)

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1:  $(m_0, m_1) \leftarrow \mathcal{A}(1^n)$                                  $\triangleright$  obtain the challenge messages,  $|m_0| = |m_1| = \ell(n)$ 
2:  $b \xleftarrow{\$} \{0, 1\}$                                            $\triangleright$  internal challenge bit, chosen by  $\mathcal{D}$ 
3:  $c \leftarrow w \oplus m_b$                                      $\triangleright$  use the challenge string as the keystream
4:  $b' \leftarrow \mathcal{A}(c)$                                      $\triangleright$  deliver the simulated challenge ciphertext
5: if  $b' = b$  then
6:   return 1                                                 $\triangleright$  guess “ $w$  is pseudorandom”
7: else
8:   return 0                                                 $\triangleright$  guess “ $w$  is uniform”
9: end if

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3 The construction and the theorem

Construction 1 (Fixed-length stream cipher). Let G be a deterministic polynomial-time function with expansion ℓ . Define $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ over the message space $\mathcal{M}_n = \{0, 1\}^{\ell(n)}$ by

$$\text{Gen}(1^n): k \xleftarrow{\$} \{0, 1\}^n; \quad \text{Enc}_k(m) = G(k) \oplus m; \quad \text{Dec}_k(c) = G(k) \oplus c.$$

Correctness is immediate: $\text{Dec}_k(\text{Enc}_k(m)) = G(k) \oplus (G(k) \oplus m) = m$. The scheme is the standard (synchronous, fixed-output-length) stream cipher: the generator expands a short key into a long *keystream* $G(k)$, and encryption is XOR of the plaintext with the keystream, exactly as in the one-time pad but with a pseudorandom pad in place of a truly random one. The key is n bits while the message is $\ell(n) > n$ bits, so—unlike the one-time pad—the scheme encrypts messages longer than its key.

Theorem 1. *If G is a pseudorandom generator with expansion ℓ , then the scheme Π of Construction 1 has indistinguishable encryptions in the presence of an eavesdropper (Definition 3). Quantitatively, for every PPT eavesdropper $\mathcal{A}$ there is a PPT distinguisher $\mathcal{D}$, running in time $t_{\mathcal{A}} + O(\ell(n))$ where $t_{\mathcal{A}}$ is the running time of $\mathcal{A}$, such that*

$$\text{Adv}_{\Pi, \mathcal{A}}^{\text{eav}}(n) \leq \text{Adv}_{G, \mathcal{D}}^{\text{prg}}(n) \quad \text{for all } n. \quad (1)$$

The asymptotic statement follows from the concrete bound (1): the right-hand side is negligible because G is a PRG, hence so is the left-hand side. We therefore prove the quantitative form.

4 Proof of Theorem 1

Strategy. We exhibit a single distinguisher $\mathcal{D}$ that turns any eavesdropper $\mathcal{A}$ into an attack on G . The distinguisher receives a challenge string $w \in \{0, 1\}^{\ell(n)}$, promised to be either $G(s)$ for a uniform seed s or a uniform string r , and it must decide which. It does so by *using w as the keystream* in a simulated run of the eavesdropping experiment against $\mathcal{A}$, and betting that w is pseudorandom exactly when $\mathcal{A}$ wins the simulated experiment. The two horns of the PRG promise then correspond to the two schemes Π (real keystream) and $\tilde{\Pi}$ (random pad, i.e. the one-time pad of Lemma 1), and the reduction reads off the theorem by comparing $\mathcal{A}$ ’s success in each.

The distinguisher is displayed as Algorithm 1. It is a PPT machine: it runs $\mathcal{A}$ once, performs one XOR of two $\ell(n)$ -bit strings, and one bit comparison, so its running time is $t_{\mathcal{A}} + O(\ell(n))$ as claimed. We now analyze $\mathcal{D}$ under each of the two input distributions of Definition 2.

Case 1: $w = G(s)$ for $s \xleftarrow{\$} \{0, 1\}^n$. We claim the view $\mathcal{D}$ presents to $\mathcal{A}$ is *identically distributed* to $\mathcal{A}$'s view in the real experiment $\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n)$. Indeed, in that experiment the key is $k \xleftarrow{\$} \{0, 1\}^n$ and the challenge ciphertext is $\text{Enc}_k(m_b) = G(k) \oplus m_b$; in the simulation the “key” is the uniform seed s underlying $w = G(s)$, and the challenge ciphertext is $w \oplus m_b = G(s) \oplus m_b$. Since s and k are both uniform on $\{0, 1\}^n$ and $\mathcal{D}$ chooses b exactly as the experiment does, the joint distribution of (m_0, m_1, b, c) —and hence of $\mathcal{A}$'s output b' —is the same in both. By construction $\mathcal{D}$ outputs 1 iff $b' = b$, i.e. iff the simulated experiment evaluates to 1. Therefore

$$\Pr_{s \xleftarrow{\$} \{0, 1\}^n} [\mathcal{D}(G(s)) = 1] = \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1]. \quad (2)$$

Case 2: $w = r$ for $r \xleftarrow{\$} \{0, 1\}^{\ell(n)}$. Now the challenge ciphertext handed to $\mathcal{A}$ is $c = r \oplus m_b$ with r uniform on $\{0, 1\}^{\ell(n)}$ and independent of b . This is exactly the challenge produced in the eavesdropping experiment against the one-time pad $\tilde{\Pi}$ of Lemma 1 (whose message space is $\{0, 1\}^{\ell(n)}$ and whose “key” is a uniform pad). Since $\mathcal{D}$ again outputs 1 iff $\mathcal{A}$ succeeds,

$$\Pr_{r \xleftarrow{\$} \{0, 1\}^{\ell(n)}} [\mathcal{D}(r) = 1] = \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{eav}}(n) = 1] = \frac{1}{2},$$

the last equality by Lemma 1. Note this holds for the very same $\mathcal{A}$ and the very same reduction; the perfect secrecy of $\tilde{\Pi}$ pins the second probability to $1/2$ irrespective of $\mathcal{A}$'s strategy.

Combining. Subtracting the two cases and using that the left-hand sides differ by at most the distinguishing advantage,

$$\begin{aligned} \mathbf{Adv}_{\Pi, \mathcal{A}}^{\text{eav}}(n) &= \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}}(n) = 1] - \frac{1}{2} \\ &= \Pr_s[\mathcal{D}(G(s)) = 1] - \Pr_r[\mathcal{D}(r) = 1] \quad (\text{by (2) and Case 2}) \\ &\leq \left| \Pr_s[\mathcal{D}(G(s)) = 1] - \Pr_r[\mathcal{D}(r) = 1] \right| = \mathbf{Adv}_{G, \mathcal{D}}^{\text{prg}}(n). \end{aligned}$$

This is exactly (1). Since G is a PRG and $\mathcal{D}$ is PPT, the right-hand side is negligible; being nonnegative and bounded above by a negligible function, $\mathbf{Adv}_{\Pi, \mathcal{A}}^{\text{eav}}(n)$ is negligible. As $\mathcal{A}$ was an arbitrary PPT eavesdropper, Π is eav-secure. $\square$

5 Remarks

Remark 1 (The hybrid, made explicit). The proof is a two-point hybrid argument in disguise. Define H_0 to be $\mathcal{A}$'s success probability against the real scheme Π and H_1 its success probability against the idealized scheme $\tilde{\Pi}$ that XORs with a truly random pad. Lemma 1 evaluates the ideal endpoint exactly, $H_1 = \frac{1}{2}$, using no computational assumption; the single distinguisher of Algorithm 1 bounds the gap $|H_0 - H_1|$ by the PRG advantage. All computational content is thus localized to the one step where $G(k)$ is swapped for a uniform string, which is the only place the assumption is used and the only source of the (negligible) security loss.

Remark 2 (Tightness). The reduction is advantage-preserving and rewinding-free: the bound (1) carries a multiplicative constant of exactly 1 and additive loss 0, and $\mathcal{D}$ invokes $\mathcal{A}$ a single time with essentially no overhead beyond one $\ell(n)$ -bit XOR. Consequently the cipher inherits the concrete security of G verbatim: an adversary spending work t and achieving eavesdropping advantage ε yields a PRG distinguisher of work $\approx t$ and advantage $\geq \varepsilon$. No security is lost in passing from the primitive to the scheme.

Remark 3 (On necessity of the assumption). Sufficiency of pseudorandomness is what the theorem asserts; a converse also holds for this construction, so the assumption is not an artifact of the proof. If G fails Definition 2, witnessed by a PPT $\mathcal{D}$ with non-negligible $\text{Adv}_{G,\mathcal{D}}^{\text{prg}}$, then Π fails Definition 3: the eavesdropper submits $m_0 = 0^{\ell(n)}$ and a uniformly random m_1 , and on receiving c runs $\mathcal{D}$ on $c \oplus m_0 = c$, guessing $b' = 0$ when $\mathcal{D}$ answers “pseudorandom.” When $b = 0$ the string fed to $\mathcal{D}$ is $G(k)$; when $b = 1$ it is $G(k) \oplus m_1$, uniform and independent of $G(k)$ because m_1 is a fresh uniform pad, so it is distributed as a truly random string. The eavesdropper’s advantage is thus $\frac{1}{2} \text{Adv}_{G,\mathcal{D}}^{\text{prg}}$, non-negligible. Hence, for this scheme, eav-security and pseudorandomness of G are equivalent.

Remark 4 (Scope: single message only). Definition 3 models a *single* ciphertext. The theorem does *not* certify the stateless scheme of Construction 1 under multiple encryptions with a fixed key: two messages encrypted under the same k produce $c_1 \oplus c_2 = m_1 \oplus m_2$, leaking the plaintext XOR and breaking any reasonable multi-message notion—precisely the two-time-pad failure. Security across many messages is recovered either by making the generator stateful (advancing the keystream so that disjoint keystream segments encrypt successive messages, as in a synchronized stream cipher) or by moving to a randomized/nonce-based construction from a pseudorandom *function*. Each requires its own definition (e.g. multiple-encryption or CPA indistinguishability) and its own reduction; neither is subsumed by Theorem 1.

Remark 5 (Uniform vs. non-uniform adversaries). The argument is agnostic to the adversary model. Read with PPT meaning uniform probabilistic polynomial time, it establishes security against uniform eavesdroppers from security of G against uniform distinguishers; read with $\mathcal{D}$ and $\mathcal{A}$ as polynomial-size circuit families and “negligible” quantified accordingly, the identical reduction establishes the non-uniform statement, since $\mathcal{D}$ merely wraps $\mathcal{A}$ with a fixed amount of extra computation and no additional nonuniform advice.